

MATH 565: Lecture 3 (01/20/2026)

Today: * more on SVM
* Taylor expansion
* local optimality conditions

Recall: SVM model:

$$\begin{aligned} \min_{\bar{w}, b, \bar{\epsilon}} \quad & J = C \sum_{i=1}^n \bar{\epsilon}_i + \frac{1}{2} \|\bar{w}\|^2 \quad (C > 0) \\ \text{s.t.} \quad & y_i (\bar{w}^T \bar{x}_i + b) \geq 1 - \bar{\epsilon}_i, \quad i=1, \dots, n \\ & \bar{\epsilon}_i \geq 0 \quad \forall i \end{aligned}$$

Let's examine the unified (main) constraints in detail:

Recall: the model tries to get $\bar{w}^T \bar{x}_i + b \geq 1$ when $y_i = +1$ and $\bar{w}^T \bar{x}_i + b \leq -1$ when $y_i = -1$.

$\bar{\epsilon}_i$: measures by how much the i^{th} sample violates well-separatedness

When $y_i = +1$, we get $\bar{w}^T \bar{x}_i + b \geq 1 - \bar{\epsilon}_i$

For instance, if $\bar{w}^T \bar{x}_i + b = 0.7$, then $\bar{\epsilon}_i \geq 0.3$ is needed;

but if $\bar{w}^T \bar{x}_i + b = 2$, then $\bar{\epsilon}_i = 0$ works, and

will be set to this value because of the $C \bar{\epsilon}_i$ term in the objective function (recall, $\bar{\epsilon}_i \geq 0$).

On the other hand, if $y_i = -1$, the constraint becomes

$$\bar{w}^T \bar{x}_i + b \leq -1 + \bar{\epsilon}_i.$$

For instance, if $\bar{w}^T \bar{x}_i + b = -3$, $\bar{\epsilon}_i = 0$ in the optimal solution.

But if $\bar{w}^T \bar{x}_i + b = 0.5$, we need $\bar{\epsilon}_i \geq 1.5$ for the constraint to hold.

Technically, we should be including the affine variable b in the regularizing term: (3.2)
 $\rightarrow$ or intercept

$$\begin{aligned} \min_{\bar{w}, b, \xi} \quad & J = C \sum_{i=1}^n \xi_i + \frac{1}{2} \left\| \begin{bmatrix} \bar{w} \\ b \end{bmatrix} \right\|^2 \\ \text{s.t.} \quad & y_i (\bar{w}^T x_i + b) \geq 1 - \xi_i, \quad i=1, \dots, n \\ & \xi_i \geq 0 \quad \forall i \end{aligned}$$

Another equivalent notation is to write $\bar{w} = [w_0 \ w_1 \ \dots \ w_d]^T$, and write the model as

$$\begin{aligned} \min_{\bar{w}, \xi} \quad & J = C \sum_{i=1}^n \xi_i + \frac{1}{2} \|\bar{w}\|^2 \\ \text{s.t.} \quad & y_i \left(\bar{w}^T \begin{bmatrix} 1 \\ \bar{x}_i \end{bmatrix} \right) \geq 1 - \xi_i, \quad i=1, \dots, n, \\ & \xi_i \geq 0 \quad \forall i. \end{aligned}$$

Note how the well-separated instances do not contribute to the objective function—irrespective of the extent of their well-separatedness. At the same time, instances that are not well-separated do incur a loss, i.e., they add to the loss function. Furthermore, the amount of this loss is more when the instance violates well-separatedness more.

We will study SVM models in detail later on...

Taylor Expansion

3.3

In one dimension, the Taylor expansion of $f(x)$ at $x=a$ is given by

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots + \frac{(x-a)^r}{r!} \left. \frac{d^r f(x)}{dx^r} \right|_{x=a} + \dots$$

When $|x-a|$ is small, we can take

$$f(x) \approx f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a)$$

as a reasonable and accurate representation of $f(x)$.

In d -dimensions, the Taylor expansion of $f(\bar{x})$ at $\bar{x}=\bar{a}$ is

$$f(\bar{x}) = f(\bar{a}) + \sum_{i=1}^d (x_i - a_i) \left[\frac{\partial f}{\partial x_i} \right] \Big|_{\bar{x}=\bar{a}} + \sum_i \sum_j \frac{(x_i - a_i)(x_j - a_j)}{2!} \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right] \Big|_{\bar{x}=\bar{a}} + \dots$$

Equivalently,

$$f(\bar{x}) = f(\bar{a}) + [\bar{x} - \bar{a}]^T \nabla f(\bar{a}) + [\bar{x} - \bar{a}]^T Hf(\bar{a}) [\bar{x} - \bar{a}] + \dots$$

for gradient $\nabla f(\bar{a})$ and Hessian $Hf(\bar{a})$ of $f(\bar{x})$ at $\bar{x}=\bar{a}$.

We will employ Taylor series approximations of functions for multiple purposes in this class - both for deriving efficient algorithms and for deriving theoretical results, i.e., proofs. Despite its somewhat simple form, the Taylor series expansion proves to be a powerful tool!

We now present local optimality conditions first in 1D and then in d -dimensions in general. We will use the Taylor series expansion to justify them.

Local Optimality Conditions in 1D

(34)

Lemma 1 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function. Then $f(x)$ is a minimum value at $x=x_0$ with respect to its immediate locality if $f'(x_0) = 0$ and $f''(x_0) > 0$.
→ (first order optimality condition)
→ (second order optimality condition)
in a small enough neighborhood of $x=x_0$

Proof Consider the Taylor expansion of f at x_0 for $x=x_0+\Delta$:
($\Delta \in \mathbb{R}$)

$$f(x_0+\Delta) \approx f(x_0) + \Delta \underbrace{f'(x_0)}_{=0} + \frac{\Delta^2}{2} \underbrace{f''(x_0)}_{>0}$$

We can take $|\Delta| \ll 1$ small enough such that

$$\left| \frac{\Delta^2}{2} f''(x_0) \right| > \left| \sum_{r=3}^{\infty} \frac{\Delta^r}{r!} f^{(r)}(x_0) \right|$$

→ the second order term dominates all other higher order terms taken together

⇒ Under first and second order optimality conditions,

$$f(x_0+\Delta) > f(x_0) \text{ for } |\Delta| \text{ small enough} \quad \square$$

The first order condition ($f'(x)=0$) is typically solved using gradient descent.

(initialization) Step 0. Start $x=x_0$ (can be chosen randomly)

do
Step k. $x_k \leftarrow x_{k-1} - \alpha f'(x_{k-1})$ (in general, $x \leftarrow x - \alpha f'(x)$)

while $|x_k - x_{k-1}| > \epsilon$.

Here $\alpha > 0$ is the learning rate (also called the step size) and $\epsilon \geq 0$ is a convergence tolerance.

Note that we are changing x by $\Delta x = -\alpha f'(x)$. In fact, we are changing x along the "steepest descent" direction — which is trivial in 1D as we have only two options ($\uparrow$ or $\downarrow$), but is nontrivial in d dimensions ($d \geq 2$) when we could have infinitely many options.

(3.5)

Our goal is to decrease $f(x)$ (we are minimizing it). Each step of gradient descent is guaranteed to decrease $f(x)$ for small values of α , since we have by Taylor expansion:

$$\begin{aligned} f(x+\delta x) &\approx f(x) + \delta x f'(x) && \text{for } |\delta x| \ll 1. \\ &= f(x) - \alpha f'(x) \cdot f'(x) \\ &= f(x) - \alpha [f'(x)]^2 \\ &< f(x) \end{aligned}$$

By the same argument, we get that $f(x+\delta x) \approx f(x)$ when $f'(x)=0$, which indicates we have converged to a local optimum.

Example

Consider $f(x) = x^2 \sin x + x$.

$$\Rightarrow f'(x) = 2x \sin(x) + x^2 \cos(x) + 1$$

$$\text{and } f''(x) = (2-x^2) \sin(x) + 4x \cos(x)$$

We explore steps of gradient descent starting at $x_0=2$ and then at $x_0=5$ using $\alpha=0.05$.

We get faster convergence for $x_0=5$ to a local minimum at $x^*=5.05$.

See course web page for Python notebook...

Local Optimality Conditions in d Dimensions

Notation We use $\bar{w}, \bar{y}$ as variables ($\bar{w} = [w_1, \dots, w_d]^T$ or $\bar{w} = [w_0, w_1, \dots, w_d]^T$) while $\bar{x}, \bar{y}$ are data in our settings of optimization for ML. The objective function is a loss function typically denoted J , e.g., $J = \frac{1}{2} \|D\bar{w} - \bar{y}\|^2 + \frac{1}{2} \|\bar{w}\|^2$ for regularized regression.

Lemma 2 Let $J: \mathbb{R}^d \rightarrow \mathbb{R}$ be a loss function. Then $J(\bar{w})$ is a minimum value at $\bar{w} = \bar{w}_0$ with respect to its immediate locality if

$$\nabla J(\bar{w}_0) = \bar{0}, \text{ i.e., } \left[\frac{\partial J}{\partial w_1} \dots \frac{\partial J}{\partial w_d} \right]^T \bigg|_{\bar{w} = \bar{w}_0} = \bar{0}, \text{ and}$$

→ (first order optimality condition)

and $HJ(\bar{w}_0) \succ 0$, i.e., the Hessian at $\bar{w} = \bar{w}_0$ is positive definite,

$$\text{(i.e., } \bar{w}^T H \bar{w} > 0 \text{ } \forall \bar{w} \in \mathbb{R}^d \setminus \{\bar{0}\}.$$

→ (second order optimality condition)

Similar to the 1D case, we can understand these conditions using the Taylor expansion of J at $\bar{w}_0$ with $\bar{w} = \bar{w}_0 + \epsilon \bar{v}$ for $\epsilon > 0$:

$$J(\bar{w}_0 + \epsilon \bar{v}) \approx J(\bar{w}_0) + \epsilon \bar{v}^T \underbrace{\nabla J(\bar{w}_0)}_{= \bar{0}} + \frac{\epsilon^2}{2} \bar{v}^T \underbrace{[HJ(\bar{w}_0)]}_{> 0 \text{ } \forall \bar{v} \neq \bar{0}} \bar{v}$$

Under the first and second order optimality conditions, we get $J(\bar{w}_0 + \epsilon \bar{v}) > J(\bar{w}_0) \text{ } \forall \bar{v} \neq \bar{0} \text{ and } \epsilon > 0.$